Songtao Wang

Ph.D. Student in Computing Science, University of Alberta

songtao2@ualberta.ca | Google Scholar | GitHub | X

RESEARCH INTERESTS

Large Language Models (LLMs) post-training and reasoning, Agentic training, and Bayesian network and Probabilistic Modeling, Reinforcement Learning, Test-Time Training,

EDUCATION

University of Alberta — Canada
Ph.D. in Computing Science — *2025 – Present*

Advisor: Prof. Xi Ye

National University of Singapore — Singapore
M.sc in Computer Engineering — *2022 – 2024*

Harbin Institute of Technology — China
B.Eng. in Automation — *2018 – 2022*

PUBLICATIONS

1. **Can Language Models Learn Failures Alone at Test Time?** *on submission*, 2026. **Current Work**
   We adapt controlled distribution reshaping in test-time training with failure-only feedback.

2. **Songtao Wang**, Quang Hieu Pham, Fangcong Yin, Xinpeng Wang, Jocelyn Qiaochu Chen, Greg Durrett, and Xi Ye. **Detecting and Suppressing Reward Hacking with Gradient Fingerprints.** *Conference on Language Modeling (COLM)*, 2026. [Paper] [Code]
   GRIFT, a gradient-based representation of reasoning trajectories for detecting reward hacking and improving rejection fine-tuning in LLM reasoning.

3. Hengguan Huang, Xing Shen, Guang-Yuan Hao, **Songtao Wang**, Lingfa Meng, Dianbo Liu, David Alejandro Duchene, Hao Wang, and Samir Bhatt.
   BayesAgent: Bayesian Agentic Reasoning Under Uncertainty via Verbalized Probabilistic Graphical Modeling. *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.
   Introduces a Bayesian agentic framework that combines LLM reasoning with probabilistic graphical models for uncertainty-aware inference and decision making.

4. Hengguan Huang*, **Songtao Wang***, Hongfu Liu, Hao Wang, and Ye Wang.
   Benchmarking Large Language Models on Communicative Medical Coaching: A Dataset and a Novel System. *Findings of the Association for Computational Linguistics: ACL*, 2024. [Paper] [Code]
   We build up **ChatCoach** a medical coaching multi-agent system and training framework, incorporating patient and coach agents, together with Generalized Chain-of-Thought for structured communication feedback.

5. Guoqing Chao, **Songtao Wang**, Shiming Yang, Chunshan Li, and Dianhui Chu.
   Incomplete Multi-View Clustering with Multiple Imputation and Ensemble Clustering. *Applied Intelligence*, 52(13):14811–14821, 2022.
   Develops a two-stage multi-view clustering approach using multiple imputation and weighted ensemble clustering to handle arbitrarily missing values across views.

TEACHING EXPERIENCE

University of Alberta — Edmonton, Canada
Teaching Assistant, Department of Computing Science — *2025 – Present*

- CMPUT267 Machine Learning, CMPUT566 Machine Learning essentials

MISCELLANEOUS

Programming Python, C/C++, ROS R, MATLAB

Hobby Go, Piano, Basketball, Football, skiing.